Prediction Review: Issues and Fixes

Oct 17, 2026 · @Lion

The predictions run and rank teams well, but five gaps let them show the wrong reasons, miss changing teams, overstate their percentages, ignore failed data checks, or over-flag small teams. This note explains each gap and the logic of the fix; code details are left out.

Summary

#	Issue	What goes wrong	Fix (logic)
1	Q12 gets credit it did not earn	On null data (Q12 has no effect), the model still passes and lists only Q12 items as risk reasons	Compare with a model without Q12; permutation test; survey-response warning; leave "what to improve" to the Outcomes page
2	No check for changing teams	A team that got a new manager can still be ranked high because of last year	Add change features and a third gate scored only on teams that changed
3	Percentages are not checked	AUC checks the order, not whether "70%" really means 70%	Calibration check; hide percentages when they are off
4	Predictions skip the health check	Ask AI refuses bad data, but the Predictions and Teams pages still show numbers	Read the health check first; hide blocked outcomes
5	Small teams make the target noisy	One leaver flips a small team from "normal" to "above average"	Smooth small teams toward the company rate

Issue 1: Q12 gets credit it did not earn

What we found. The team turnover model (model A) uses three inputs: each Q12 item's team average, department and team size. On the null test data, where Q12 has no effect on any outcome, the model still passed (AUC 0.63) and listed only Q12 items as risk

reasons. The program only counts Q12 items when it explains a team's risk, so all the credit went to Q12.

Where the accuracy really came from (null data):

Source	Adds	Evidence
Random guessing	0.50	Baseline
Department and team size	+0.11 (to 0.61)	Model B, with department and size only, already reaches 0.61; Sales has higher turnover by design (trap X1)
Survey-response pattern X7, carried in through the Q12 columns	+0.02 (to 0.63)	The A − B gap. Across 21 null datasets it averages 0.034; with X7 switched off it drops to about 0.001
Q12 itself	0	Q12 has no effect in the null design

Why the scores mislead. People who will leave, and are less engaged, answer the survey less often. The leavers who do answer are the more satisfied ones, so teams with more future leavers end up with *higher* average scores. On null data, teams with higher scores had more turnover, the opposite of intuition. Read as "Q12 drives turnover," this would lead to wrong advice. This can also happen in a real company, and it can sit on top of a real Q12 signal; accuracy alone cannot tell them apart.

Fix: three layers.

1. **Does Q12 add predictive information?** Compare model A with model B (department and size only) on the same test teams. Then shuffle the Q12 scores between teams within each department 200 times, refit A each time, and record the A − B gap. The real gap must beat 95% of the shuffled gaps. Passing allows Q12 items to be shown as **warning signs**, never as causes. Null data still passes this step, because the X7 signal is real, so layers 2 and 3 are needed.
2. **Warn about the survey-response pattern.** When the health check `NR_LEAVERS` shows WARN (leavers answered less often than stayers), the prediction pages show: "Scores for some teams may look better than reality. People who later left were less likely to answer the survey (60% vs 86% of those who stayed), so the answers come mostly from people who are more satisfied."

3. **Only the Outcomes page says which Q12 items to improve.** Its analysis controls for confounders and corrects for multiple tests, and it correctly finds no link on null data. Prediction warning signs decide which teams to check in with first; they are never the basis for "improve this question."

Issue 2: No check for teams whose situation changed

What we found. A prediction is shown only if it passes two gates: it beats random guessing, and it beats "rank teams by last year's turnover." Both are scored across all teams at once, so we never learn whether the model notices change. A team that got a better manager, or changed how it works, may have had high turnover last year and still be fine next year.

Fix, part 1: let the model see change. Besides this year's scores, add each item's change from last year, the change in the overall Q12 average, and, when the data has a `manager_id`, whether the manager changed.

Fix, part 2: a third gate, scored only on teams that changed (for example, Q12 average moved by more than 0.3 points, or a new manager). In the test year, compare the model with "rank by last year":

Measure	Question it answers	Proposed bar
AUC on changed teams	Among changed teams, does the model rank the ones that really had high turnover first?	At least 0.05 above "rank by last year"
Turnaround recall (sensitivity)	Of teams that were High last year and really improved, how many did the model move out of High?	Clearly above "rank by last year" (which is always 0%)
False turnaround rate	Of teams the model moved out of High, how many still had above-normal turnover?	At most 20%
Number of changed teams	Is there enough data to judge?	At least 30

If the gate passes, the team list can show "risk lower than last year: Q05 +0.6, Q07 +0.5 (new manager)." If it fails or there are too few changed teams, only the overall ranking is shown. All bars are proposals for the team to agree on.

Issue 3: Risk percentages are not checked

What we found. AUC only checks the order: are the teams that really had high turnover ranked first? It does not check the numbers. A model can rank perfectly and still say "80%" when only 40% of such teams have above-normal turnover. The Teams page calls the risk an "estimated chance," so the number should be right. The employee model already has this check; the team model does not.

Fix: a calibration check on the test year. Sort teams by predicted risk and cut them into equal groups. For each group, compare the average predicted chance with the share of teams that actually had above-normal turnover. Example from the seed2 test set (3,789 teams):

Group	Teams	Predicted (avg)	Actual
1 (lowest)	758	28%	30%
2	758	36%	34%
3	757	42%	43%
4	758	49%	44%
5 (highest)	758	70%	67%

Note that "70%" means a 70% chance that the team's turnover is above the company rate, not that 70% of its people leave.

Rules (proposed):

- A group is off only if its gap is both larger than 10 points and larger than chance alone would explain (about $2 \times \sqrt{p \times (1 - p) \div n}$, with p the group's predicted chance and n its teams).
- Each group needs at least 100 teams: 500+ teams → 5 groups; 300–499 → 3 groups; under 300 → "insufficient." Only teams with 5+ people count.
- No later year with known results yet → "pending."

Result	Risk bands (High / Elevated)	Percentages
Good	Shown	Shown, with "when the model said about 70%, 67% happened"
Off	Shown (the ranking rests on AUC)	Hidden: "the ranking is reliable, but the percentages may be too high or too low"
Insufficient or pending	Shown	Hidden, with the reason

Issue 4: Predictions skip the health check

What we found. In v3, Ask AI refuses to interpret data that fails the health check, for example duplicate employees, scores outside 1–5, or outcome dates before the survey. The Predictions and Teams pages compute straight from the uploaded file and never look at the health check. With a file whose turnover dates are wrong, Ask AI says "turnover analysis is blocked," while the Predictions page still shows "about 420 expected leavers next year." The two pages contradict each other, and a manager may trust numbers built on bad data.

Fix: every page reads the health check first.

Health check result	Prediction pages
All checks pass	Shown as usual
One outcome blocked (for example turnover)	That outcome's predictions hidden with a short reason; the others shown
Everything blocked	No predictions; "This file has data errors. See the Data page to fix them."

Warnings carry over too: when `NR_LEAVERS` is WARN, the survey-response note from Issue 1 appears on these pages. The same rule applies to the facts Ask AI uses when it answers questions about next year's risk.

Issue 5: Small teams make the target noisy

What we found. The model learns from a 1/0 label: is the team's turnover above the company rate? For a small team, one person leaving or staying flips that label. With a company rate of 15%:

Team size	0 leave	1 leaves	2 leave
6	0%, below	17%, above	33%, above
10	0%, below	10%, below	20%, above
50	0%, below	2%, below	4%, below

Small teams land at the extremes, so their label often reflects one person's decision, not the team. The model then learns noise, accuracy drops, and the High list leans toward small teams.

Fix: smooth small teams toward the company rate before setting the label: smoothed rate = (leavers + k × company rate) ÷ (team size + k). With k = 20, a 6-person team with 1 leaver moves from 17% to 15%, while a 50-person team with 10 leavers barely moves (20% to 19%). A team is labelled "above" only if its smoothed rate is at least 3 points above the company rate. k and the 3-point margin will be tuned on the synthetic data, where each team's true risk is known.

A larger option for later: predict the number of leavers per team instead of a 1/0 label.